LabPulse

Software Requirements Specification

Real-Time Lab Monitoring, Geofenced Attendance and Asset Booking System

Karunya Institute of Technology and Sciences

Division of Computer Science and Engineering

Course: 23CS2048 – Web Technology (3 IA Project)

Reference: Circular KITS/CSE/CIR/2026/163, the client's GeoFence Proposal and the supervisor's briefing

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What changed in version 1.4

After version 1.3 the supervisor briefed the teams (two voice messages). Version 1.4 adds everything in that briefing. The first deployment is the **Intel Unnati IoT Lab**; nothing is tied to one room, so the same app serves every lab in the department.

Supervisor asked for	What LabPulse now does	Requirement
See from home how many students are in the lab, which systems they use and what projects they work on	Check-in records the system (lab PC, GPU node, 3D printer, drone or own laptop) and, for extra hours, the project. HOD, Coordinator and staff see a live “who’s in the lab” list: name, register number, system, project, time in	FR-38
Assets such as 3D printers, drones and computers with graphics cards, used by timeline, with a queue to reserve a spot	Catalog includes them. Students reserve a time slot; if it is taken they join a waitlist and are emailed when it frees up. Staff start the session only when the student is checked in. No-shows release the slot	FR-40, FR-48
Record the project, its timeline and its outcome	Projects with title, members, start and end date, a work log built from each visit (system, hours, “what I did today”), and a submitted outcome with a link	FR-39
Given a register number: how many days and how long in the lab	Lookup by register number: days in lab, total hours, visits by day, systems, projects	FR-41
Given a system: how many people used it and for what	Lookup by system: number of people, uses, hours, and each use with person, purpose and project	FR-42
Regular lab (e.g. Web Technology at 11 am): faculty find who is missing	Roster shows Present · Pending · Missing; faculty are emailed the missing list after the session; CSV download	FR-43
Google Sheets storage with periodic backups and multiple sheets so one sheet is not overloaded	Nightly per-day summaries, monthly archiving of old rows to a yearly archive spreadsheet, nightly backup copies with retention	FR-44, FR-45
Encrypt crucial data such as phone numbers, and validate input	Phone numbers stored encrypted with tamper detection, shown only to roles that need them; stricter validation of register numbers, names and all input on the server	FR-46, FR-47
Roles: student (sees own hours), coordinator (add/remove facilities and students), HOD (sees everything), super admin (manages roles)	Six roles: Student, Faculty, Lab staff, Coordinator, HOD (read-only), Super Admin	FR-37

Changes in version 1.3

Version 1.2 was our plan. Version 1.3 aligns that plan line by line with the client’s **GeoFence Proposal** and records what has now been built and tested. The proposal is treated as the source of truth; where our plan differed, the plan has been changed.

Area	Version 1.2	Version 1.3 (matches the proposal)
Live occupancy counter	Check-ins pending review counted as inside	Counter increments only on a **verified** sign-in; check-ins awaiting review are shown beside it (“12 inside · 2 pending”)

Area	Version 1.2	Version 1.3 (matches the proposal)
Reservation gating	Any active visit could request equipment	Requires an active, verified CHECKED_IN status
Entry reasons	Courses and events only from the timetable list	Timetable list, plus a “not listed” form with exactly the proposal’s fields (course code, course name, session hours; event name, coordinator, expected duration)
Research equipment	Free text	Chosen from the lab’s own hardware catalog, plus optional free text
Mock location	“Flag” suspected fake GPS	**Reject** the check-in when the browser reports a mock location
Daily digest	Single peak hour	Peak occupancy **hours** (three busiest)
Export formats	CSV	CSV **and Excel (.xlsx)**, both ERP-ready
Backend endpoints	doGet	doGet **and doPost**, sharing one role-checked API
Status	Requirements only; nothing built	All P0 and P1 requirements built; 48 automated checks pass. Not yet deployed to Google.

1. Introduction

1.1 What we are building

Today a specialised lab runs on a paper register and memory. Someone signs in for a friend, nobody knows how many people are inside, and a Raspberry Pi borrowed on Monday is forgotten by Friday. LabPulse replaces this with a Google Apps Script web app that verifies who is physically in the lab, shows live occupancy, manages high-value hardware, and produces the reports the department needs.

As the proposal requires, it is built 100% on Google Apps Script and Google Workspace at zero server cost. This document describes the system architecture, the geofencing workflow and the dashboard designs, and maps every line of the proposal to a requirement (Section 1.3).

1.2 Approach and added value

The design philosophy is to make correct behaviour the easiest behaviour: if checking in takes about ten seconds and misuse takes more effort than simply being present, users follow the process. Every requirement in the proposal is met (priority P0). On top of that we add four ideas. They use only Google Apps Script, Sheets and Workspace, and none of them changes what the proposal asks for.

Added idea	What it does	Why
Presence Confidence	Each check-in is graded Green, Amber or Red from QR validity, distance, GPS accuracy, travel speed and the mock-location signal.	GPS drifts indoors. Rejecting an honest student is unfair; accepting everything is unsafe. Amber goes to one-tap faculty or staff review and is only counted once confirmed.
Consume-and-Refresh QR	When a QR is scanned, the monitor shows a fresh one within about 1.5 seconds.	The proposal requires single-use tokens. Implemented naively, only one student could check in per rotation.

Added idea	What it does	Why
Self-Healing Occupancy	Forgotten visits close automatically and are labelled; the counter is recounted from records every 5 minutes.	A live counter is only useful if it is accurate.
Live Lab Map	A floor-plan view of which workstations, GPU nodes and kits are free or in use.	Answers the question students ask most: is anything free right now?

An optional extra, the **Lab Passport** (a one-page PDF of a student's verified hours, courses and equipment handled), is also built.

Limits of the claims. The Lab Map is built from application records, not physical sensors. Presence Confidence reduces misuse but does not make it impossible, and browsers do not expose a mock-location signal on every phone. The system states these limits openly.

1.3 Traceability to the client proposal

Every line of the proposal, the requirement that covers it, and its status. "Built and tested" means automated tests pass; "verify on deployment" marks features whose logic is tested but which call Google Docs, Drive or Gmail, which only run once deployed.

Prop. §	Proposal requirement	Covered by	Status
A	Domain lock to @karunya.edu / @karunya.edu.in	FR-01	Built and tested
A	Profile: name, roll no. / employee ID, Karunya email, mobile	FR-02	Built and tested
A	RBAC – Student: scan QR, GPS check-in/out, live counter, visit reason, request bookings, borrowing history	FR-03, 05–10, 14, 21, 25	Built and tested
A	RBAC – Faculty: live roster, verify attendance, export session logs	FR-27	Built and tested
A	RBAC – Lab In-charge / Staff: activity stream, live occupancy, inventory add/edit/remove, issue/return handshake	FR-22, 24, 26	Built and tested
A	RBAC – Super Admin: GPS coordinates, radius, hours, role permissions, audit logs	FR-35, 36	Built and tested
B	Dynamic TOTP-style QR on a lab monitor, rotating every 5–10 s	FR-05	Built and tested
B	GPS geofence with Haversine, 15–20 m	FR-07	Built and tested
B	Categorised entry: course (code, name, hours), event (name, coordinator, duration), research (purpose, hardware)	FR-09	Built and tested
B	Live counter: increment on verified sign-in, decrement on sign-out, on all dashboards	FR-14	Built and tested
C	Catalog: GPU (RTX 4090 / A100), IoT (ESP32, Raspberry Pi, LoRaWAN), sensors (ultrasonic, thermal, LiDAR, gas)	FR-19	Built and tested
C	Public availability catalog for students	FR-20	Built and tested
C	Reservation gated on active CHECKED_IN status	FR-21	Built and tested
C	Staff handshake UI to approve checkout and mark returns	FR-22	Built and tested
C	Overdue tracking and automated reminders	FR-23	Built and tested
2	Single-use token invalidation in CacheService	FR-06	Built and tested
2	LockService during check-ins	FR-11	Built and tested

Prop. §	Proposal requirement	Covered by	Status
2	Impossible-travel velocity check: flag and block	FR-12	Built and tested
2	Mock-location detection: reject emulated GPS	FR-13	Built and tested
2	Privacy: store is_valid_geofence and timestamp; discard latitude/longitude	FR-08	Built and tested
2	Append-only audit of manual entries, role changes, inventory overrides	FR-04	Built and tested
3	Daily digest: unique visits, peak hours, hourly distribution, pending/unreturned items	FR-29	Built and tested
3	Monthly audit: course-wise student-hours, GPU utilisation, sensor turnover	FR-30	Built and tested
3	Custom start–end date range export	FR-31	Built and tested
3	PDF from a Google Docs template, saved to Drive	FR-32	Built; verify on deployment
3	CSV / Excel for ERP import	FR-32	Built; verify on deployment
3	Daily end-of-day email to the Lab In-charge via GmailApp	FR-33	Built; verify on deployment
4	Google Apps Script V8, doGet / doPost	§3.1	Built and tested
4	HtmlService single-page app (HTML5, Tailwind CSS, JavaScript)	§3.1, §6	Built and tested
4	Google Sheets database with structured tabs	§7	Built and tested
4	DriveApp, DocumentApp, GmailApp	FR-32, 33	Built; verify on deployment
4	PropertiesService for geofence limits and constants	FR-35	Built and tested
–	Presence Confidence, Consume-and-Refresh, Self-Healing Occupancy, Lab Map, Lab Passport	FR-16, 17, 18, 28, 34	Added value (beyond proposal)

2. Users and Roles

Access is restricted to institutional Google accounts (@karunya.edu and @karunya.edu.in). Each user records name, roll number or employee ID, institutional email and mobile number.

Role	What they can do
Student	Scan the QR and check in or out with GPS, see the live lab counter and their own hours and days, choose a visit reason, system and project, manage projects (timeline, outcome), reserve equipment slots, request items, see borrowing and visit history
Faculty	See the live roster for their courses or events (Present · Pending review · Missing), confirm flagged check-ins, mark present or absent with a reason, export session logs, receive the missing list by email
Lab Staff	Run the lab monitor, watch live activity, hand over and take back equipment, start and finish reserved sessions, make supervised manual entries, review research check-ins

Role	What they can do
Coordinator	Everything Lab Staff can do, plus add, edit and remove facilities (inventory), add and remove students, manage courses and events, see live usage, lookups and reports
HOD	Read-only view of every lab: who is in it, on which system and project, rosters, projects, lookups by register number or system, reports and exports. Cannot change anything
Super Admin	Manage every role and permission, configure GPS coordinates, geofence radius, operating hours and system settings, inspect audit logs, run maintenance jobs

3. System Architecture

3.1 Layers

Everything lives inside Google Workspace, with no external server.

Layer	What it is	What it does
Browser	Single-page responsive web app (HTML5, Tailwind CSS, JavaScript) served by HtmlService	Role dashboards, the lab-monitor QR page, camera link and GPS access
Application server	Google Apps Script V8: **doGet** (pages), **doPost** (JSON API) and google.script.run, all through one role-checked dispatcher	All business rules: token checks, geofence, roles, bookings, reports
Guard layer	CacheService, LockService, PropertiesService	Single-use token memory, protection from simultaneous check-ins, settings and secrets
Data	Google Sheets	Every record (Section 7)
Output	Google Docs, Drive, GmailApp	PDF reports, CSV and Excel exports, daily summary email

3.2 Design principle: server-side trust

The browser is never trusted. The server decides.

The phone only reports what it observed (a QR value, a location and its accuracy). Every decision is made in Apps Script. Hiding a button is not security; a server-side role check is. In the built system, one route table lists which roles may call each action, and every request is checked against it.

3.3 How a request travels

1. The user opens the web app URL and Google sign-in identifies them.
2. The server checks the email domain and looks up the user’s role.
3. The screen shows only what that role may use.
4. Every action goes to the server, which re-checks identity and role, validates input, takes a lock if it writes, writes, and replies with a clear result. Sensitive actions are also appended to the audit log.

3.4 Background jobs

Job	How often	What it does
Occupancy reconciliation	Every 5 minutes	Recounts verified and pending occupancy from active visits and fixes any drift
Auto-close sweep	Every 10 minutes	Closes visits past their duration plus grace, or past closing time
Overdue check	Hourly	Marks late loans, emails reminders to the borrower and staff, expires stale requests
Daily digest	Hourly check; sends once, 30 min after each lab closes	Builds the report, saves the PDF to Drive, emails the Lab In-charge; nightly spreadsheet backup
Monthly audit	1st of each month	Generates last month's institutional report

3.5 Ready for every lab

Nothing is hard-coded to one room. Every record carries a lab ID, and each lab has its own coordinates, radius, hours and monitor QR. Adding a lab is a settings change, not a rewrite.

4. Geofencing and Attendance Workflow

4.1 The student's view (target: under 10 seconds)

1. Point the phone's normal camera at the QR on the lab monitor and tap the link. There is no app to install; LabPulse opens, signed in with Google.
2. Allow location when asked.
3. Pick one reason: **Regular Lab Course** (course code, course name, session duration in hours), **Event / Workshop** (event name, event coordinator name, expected duration) or **Individual Research / Self-Work** (purpose of visit, specific hardware/equipment required). Courses and events running now are offered from the timetable so their details fill in automatically; if one is missing, the student types those same fields. Research hardware is chosen from the lab's own catalog, with free text for anything else.
4. See a clear result: **Checked in**, **Checked in — pending review**, or **Not checked in, and here is why**.

4.2 What the server does behind the scenes

Step	Check	If it fails
1	Caller is a signed-in institutional user with an active profile	Reject; log event
2	Lab is open (inside operating hours)	Reject with a clear message; staff can make a logged override
3	QR token has a valid signature and is within its time window	Reject as expired or invalid; log event
4	Token has not been used already (checked and marked under a lock)	Reject as reused; log security event
5	User has no other active visit in this lab	Show the existing visit instead of creating another
6	Distance from the lab centre (Haversine) is inside the configured radius, 15–20 m	Reject as outside the lab (Amber if only just outside, within GPS uncertainty)
7	Reported GPS accuracy is reasonable for the radius	Downgrade to Amber

Step	Check	If it fails
8	Travel speed since the previous attempt is plausible	Flag (Amber) above the walking limit; block (Red) when impossible
9	Browser mock-location signal, where one exists	**Reject** (Red). A missing signal is not treated as proof of honesty
10	Visit reason details are complete; session is in the timetable	Ask the student to complete them; an unlisted session goes to Amber for staff confirmation
11	Write the visit, the audit event and the occupancy update	Reply with the result

Latitude and longitude exist only in server memory while steps 6 to 8 run. What gets stored is the verdict (`is_valid_geofence`), the confidence level and a timestamp. The travel check keeps only the previous attempt's *distance from the lab centre*, never a position, and uses the triangle inequality to bound how far the user moved. Our own logging never prints the coordinates.

4.3 Presence Confidence

Outcome	When	What happens
Green	Fresh valid QR, inside radius, good GPS accuracy, normal travel history, session in the timetable	Checked in and counted immediately
Amber	Valid QR and near the lab, but accuracy is poor, travel is unusual, or the session is not in the timetable	Checked in as *pending review*; shown as "pending" beside the counter. Faculty (timetabled sessions) or staff (research and unlisted sessions) confirm or decline with one tap. Confirmed visits join the counter; declined ones are voided and logged
Red	Reused or invalid QR, clearly outside the radius, impossible travel speed, mock location detected	Not checked in. The reason is shown and the event is logged

Thresholds (radius, accuracy limit, speed limits) are settings, not code, so admins can tune them after testing in the real lab.

4.4 Consume-and-Refresh QR

The lab monitor shows a QR containing only the lab ID, a time-window number, a random nonce and an HMAC-SHA256 signature made with a secret key that stays in Script Properties. **No reusable secret is ever inside the QR.**

- The QR rotates every **8 seconds** (configurable within the required 5–10 seconds).
- When a scan succeeds, the server marks the token as used in CacheService and the monitor swaps to a new QR within about 1.5 seconds.
- Opening the link consumes the token and gives that user a 5-minute server-side ticket to finish the form, so a slow location fix does not need a rescan.
- A screenshot is therefore useless: it is either expired or already consumed by whoever scanned it first.

Quota note. The monitor's refresh reads only the cache and Script Properties, never a Sheet, and slows to once a minute outside operating hours. **Throughput note.** About one student every 2–3 seconds per monitor; a class of 60 needs a couple of minutes. Faster options (a second monitor, a faculty-opened entry window) need supervisor approval before we change the single-use rule.

4.5 Check-out and the live counter

Students check out with one tap. The counter shows **verified** people inside: it goes up on a verified sign-in (Green, staff manual entry, or a confirmed Amber) and down on sign-out, decline or auto-close. Check-ins awaiting review appear beside it, so the number on screen is never inflated and nothing is hidden. If a student forgets to check out, a course or event visit closes at its stated duration plus a grace period and a self-work visit closes at lab closing time; auto-closed visits are labelled in data and reports. Every 5 minutes both counts are recalculated from active visit records.

4.6 When things go wrong

Situation	What the user sees	What the system does
Location permission denied	A clear explanation and how to enable it	Nothing is written. Staff can make a supervised manual entry, which is audit-logged
Network drops mid-request	“Checking… please wait”, then the final result	Each request carries a unique ID, so a retry never creates a duplicate visit
QR monitor is down	The monitor hides the code and shows *Offline*	No attendance is ever created silently. Staff follow the manual fallback
Session not in the timetable	A form for the session details	Check-in recorded as pending; lab staff confirm

5. Functional Requirements

Priority key: **P0** = required by the circular and the client proposal. **P1** = our added ideas. **P2** = optional. Status reflects the build as of this version.

5.1 Access and roles

ID	Requirement	Priority	Status
FR-01	Only @karunya.edu and @karunya.edu.in Google accounts can enter. Everyone else is rejected on the server.	P0	Built, tested
FR-02	First sign-in captures name, roll number or employee ID and mobile number; the email comes from the Google account.	P0	Built, tested
FR-03	Student, Faculty, Lab Staff and Super Admin roles are enforced on the server for every request.	P0	Built, tested
FR-04	Role changes, manual administrative entries and inventory overrides are written to an append-only audit log.	P0	Built, tested

5.2 Attendance

ID	Requirement	Priority	Status
FR-05	A lab-monitor page shows a time-based (TOTP-style) QR that rotates every 5–10 seconds.	P0	Built, tested
FR-06	Tokens are signed, expire quickly, and are invalidated on first use through CacheService.	P0	Built, tested
FR-07	Check-in measures distance from the lab with the Haversine formula against a 15–20 m configurable radius.	P0	Built, tested

ID	Requirement	Priority	Status
FR-08	Raw latitude and longitude are discarded after evaluation. Only is_valid_geofence and a timestamp are stored.	P0	Built, tested
FR-09	The student must choose one entry reason with its required fields: course (code, name, session hours), event (name, coordinator, expected duration) or research (purpose, hardware from the lab catalog). Timetabled sessions fill in automatically; unlisted ones are typed and go to staff review.	P0	Built, tested
FR-10	Check-in creates one active visit; a second active visit for the same user and lab is blocked; check-out closes it.	P0	Built, tested
FR-11	Check-in is wrapped in LockService so simultaneous arrivals cannot corrupt the data.	P0	Built, tested
FR-12	Travel speed between consecutive attempts is computed; unusual speed is flagged and impossible speed is blocked.	P0	Built, tested
FR-13	Mock-location flags are read where the browser exposes them, and a check-in reporting a mock location is **rejected**.	P0	Built, tested
FR-14	The live occupancy count appears on every dashboard and the monitor, refreshes every 5–10 seconds, counts only verified active visits, and shows pending check-ins separately.	P0	Built, tested
FR-15	Check-in is accepted only during the lab’s operating hours; staff can override with a logged reason.	P0	Built, tested
FR-16	Presence Confidence: each check-in is Green, Amber or Red with a stored reason; Amber goes to a review queue.	P1	Built, tested
FR-17	Consume-and-Refresh: the monitor shows a new QR within about two seconds of a successful scan.	P1	Built, tested
FR-18	Self-Healing Occupancy: forgotten visits are auto-closed and labelled, and occupancy is reconciled on a timer.	P1	Built, tested

5.3 Assets and booking

ID	Requirement	Priority	Status
FR-19	The catalog holds GPU workstations (RTX 4090 / A100), IoT hardware (ESP32, Raspberry Pi, LoRaWAN gateways) and sensors (ultrasonic, thermal, LiDAR, gas). Each item has a unique ID, category, status and condition.	P0	Built, tested
FR-20	Students see a public availability list (Available / Reserved / Issued) with no borrower details.	P0	Built, tested
FR-21	A booking request is accepted only from a student whose status is currently CHECKED_IN with a verified visit to that lab.	P0	Built, tested
FR-22	Staff confirm the physical handover and return. The system records who issued, who borrowed, issue time, due time, return time and condition.	P0	Built, tested
FR-23	Overdue items are flagged automatically. Borrowers get reminder emails and staff see an overdue list.	P0	Built, tested
FR-24	Staff can add, edit or remove inventory records, and each action is audit-logged.	P0	Built, tested
FR-25	Students can see their own borrowing history and visit history.	P0	Built, tested

5.4 Live monitoring

ID	Requirement	Priority	Status
FR-26	Staff see a live activity stream (check-ins, check-outs, issues, returns, security events) and current occupancy.	P0	Built, tested
FR-27	Faculty see the roster for their active course or event (Present · Pending review · Not yet arrived), verify attendance live (including manual present or absent with a reason), and export the session log.	P0	Built, tested
FR-28	Live Lab Map: a floor plan showing workstation and asset status from stored records, labelled “based on app records, not physical sensors”.	P1	Built, tested

5.5 Reports and exports

ID	Requirement	Priority	Status
FR-29	Daily operational digest: unique visits, peak occupancy hours, hourly attendance distribution, pending or unreturned items.	P0	Built, tested
FR-30	Monthly institutional audit: course-wise student-hours, GPU workstation utilisation, sensor turnover frequency.	P0	Built, tested
FR-31	Custom report and export for any start–end date range (multi-day workshops, hackathons, assessment weeks).	P0	Built, tested
FR-32	A Google Docs template is filled and saved as a PDF in Drive. CSV and Excel (.xlsx) exports are ERP-ready: one flat table, fixed column names, ISO dates.	P0	Built; verify on deployment
FR-33	A scheduled end-of-day summary email goes to the Lab In-charge through GmailApp.	P0	Built; verify on deployment
FR-34	Lab Passport: a one-page PDF of a student’s verified hours, courses and equipment handled.	P2	Built; verify on deployment

5.6 Administration

ID	Requirement	Priority	Status
FR-35	Super Admin sets lab coordinates, radius, operating hours and system constants (stored in PropertiesService), per lab.	P0	Built, tested
FR-36	Super Admin manages role permissions and views the audit log.	P0	Built, tested

5.8 Supervisor briefing additions (version 1.4)

ID	Requirement	Priority	Status
FR-37	Six roles: Student, Faculty, Lab Staff, Coordinator, HOD (read-only) and Super Admin, enforced on the server for every request. Coordinators manage student accounts only; only the Super Admin changes other roles.	P0	Built, tested

ID	Requirement	Priority	Status
FR-38	Check-in records the system used (lab computer, GPU node, 3D printer, drone, or own laptop) and the project for extra-hours visits. A live view lists everyone inside with register number, system, project and time in; systems in use show on the Lab Map.	P0	Built, tested
FR-39	Projects have a title, description, members, start and end date (timeline), status and a submitted outcome with an optional link. Check-out asks what was done; each project shows a work log and total hours.	P0	Built, tested
FR-40	Students reserve equipment for a time slot inside opening hours (length and advance limits are settings). Overlapping requests join a first-come waitlist with a visible position; cancellations and no-shows promote the next person, who is emailed. A session starts only when the student is checked in.	P0	Built, tested
FR-41	Lookup by register number or email: days in the lab, total hours, visits by day, systems used and projects. Students see their own totals.	P0	Built, tested
FR-42	Lookup by system or item: number of different users, number of uses, total hours, and each use with person, purpose or project.	P0	Built, tested
FR-43	For timetabled sessions, the roster marks enrolled students who did not attend as Missing once the session ends; the session faculty is emailed the list once; it can be downloaded as CSV.	P0	Built, tested
FR-44	Nightly per-person and per-item daily summaries; monthly, closed rows older than a configurable number of months move to a yearly archive spreadsheet without changing totals.	P0	Built, tested
FR-45	Nightly backup copy of the spreadsheet, keeping the last N daily copies and one per older month.	P0	Built; verify on deployment
FR-46	Phone numbers are stored encrypted with integrity protection; only Super Admin, Coordinator and staff following up a loan see the full number; others see a masked form; exports never include it.	P0	Built, tested
FR-47	Server-side validation of register numbers (e.g. URK24CS1249), names, phone numbers, emails, dates, slot times and text lengths; control characters are stripped.	P0	Built, tested
FR-48	The catalog includes 3D printers, drones and lab computers with graphics cards, in addition to the proposal’s GPU nodes, IoT boards and sensors.	P0	Built, tested

5.7 How the report numbers are calculated

Metric	Definition
Unique visits	Different people with at least one verified visit that day
Peak occupancy hours	The three hours with the highest number of verified people inside at once
Hourly distribution	For each hour, the number of different people present at any time in that hour
Course-wise student-hours	Sum of (check-out minus check-in) for verified visits tagged to that course, clipped to the period; auto-closed visits count only up to the time they were closed; unlisted sessions are reported under the code and name the student entered
GPU workstation utilisation	Hours reserved or issued, within operating hours, divided by operating hours available in the period
Sensor turnover frequency	Issue-and-return cycles for a sensor type divided by the number of units of that type

6. Dashboard Designs

Every dashboard shows the live counter (“N inside · M pending”) and handles loading, empty, error and permission-denied states.

6.1 Student (phone-first)

- **Top:** status card: *Not checked in* with a Scan button, or *Checked in · 1h 12m* with a Check-out button and, if pending, a note that a review is waiting.
- **Middle:** “Free right now” chips (for example *2 GPU nodes · 5 ESP32 · 1 LiDAR*) and a request button that stays disabled until the check-in is verified, with a note explaining why; the Live Lab Map.
- **Bottom:** borrowed items with due dates, recent visits; full history and Lab Passport on a second tab.

6.2 Faculty

- **Left:** today’s sessions and the active roster, split into *Present · Pending review · Not yet arrived*, with manual present/absent.
- **Right:** pending reviews with Confirm and Decline buttons and the reason each was flagged.
- **Footer:** export the session log (CSV or PDF).

6.3 Lab Staff

- **Top strip:** inside now (+ pending), items out, overdue count, pending requests, pending reviews.
- **Left:** live activity stream. **Centre:** Live Lab Map and who is inside. **Right:** the handshake panel to confirm an issue with a due time or mark a return with its condition.
- **Tabs:** Reviews, Inventory, Courses & events, Reports (daily, monthly, date range, CSV and Excel export).

6.4 Super Admin

- **Settings:** lab location (with a *use my current location* helper), radius, operating hours, thresholds, monitor link, background jobs.
- **People:** role management. **Permissions:** grant delegable actions to more roles.
- **Audit:** security events and the audit log, filterable by actor, action and date.

6.5 Lab Monitor (wall display)

A clean full-screen page with a large QR, a countdown ring, the verified count with pending beside it, and a visible *Online / Offline* indicator. When offline it hides the code rather than show a stale one.

7. Data Design

Google Sheets is the database: one tab per kind of record, unique IDs, fixed status words, and a lab ID on every row.

Tab	Holds	Key fields
Users	Everyone with access	user_id, email, name, roll/employee ID, mobile, role, active

Tab	Holds	Key fields
Config	Per-lab settings (mirrored in Script Properties)	lab_id, coordinates, radius, hours, thresholds; secrets stay in Script Properties
CoursesEvents	Sessions people can check in to	id, type, code, name, coordinator, days, start/end time and date, session hours
Enrollments	Expected attendees per session	session_id, user_email
Visits	Attendance history	visit_id, user_id, lab_id, entry_type, course_or_event_id, purpose, details, check_in_at, check_out_at, is_valid_geofence, confidence, review_status, close_type, status
ActiveVisits	People inside now, for fast checks	visit_id, user_id, lab_id, check_in_at, review_status
AttendanceMarks	Faculty manual present/absent	session_id, date, user_email, mark, reason, marked_by
Assets	Inventory	asset_id, name, category, type, status, condition, location, map position
Requests	Booking requests	request_id, asset_id, requester_id, window, status, decided_by, decided_at
Loans	Issued items	loan_id, asset_id, borrower_id, issued_by, issued_at, due_at, returned_at, return_condition, status
AuditLog	Append-only history	event_id, timestamp, actor, role, action, target, outcome, reason
Projects	Student projects	project_id, title, description, owner, members, start_date, end_date, status, outcome, outcome_link
Bookings	Time-slot reservations and queue	booking_id, asset_id, user, project_id, start_at, end_at, purpose, status (BOOKED, WAITLIST, IN_USE, DONE, CANCELLED, NO_SHOW)
UsageDaily	Nightly summary per person per day	date, lab_id, email, visits, minutes, first in, last out, projects, systems
AssetUsageDaily	Nightly summary per item per day	date, lab_id, asset_id, uses, minutes, users, purposes
(Archive YYYY)	Separate yearly spreadsheet	Same tabs as above for rows moved out of the live spreadsheet

Rules. IDs are unique and required fields are validated before every write. An asset with an active loan cannot be given to someone else. Occupancy is always calculated from active visits, never typed in, and ActiveVisits is rebuilt from Visits during reconciliation. Audit rows are only added by the application; because the sheet owner can still edit by hand, sharing settings are kept private. The script time zone is Asia/Kolkata.

8. Security and Privacy

Attempt	Our defence	How we test it
Sharing a screenshot of the QR	Short expiry, signed token, single-use consumption	Scan the same token twice; the second must fail (AT-03)
Checking in from the hostel	Geofence, accuracy check, travel-speed check	Send coordinates outside the radius; expect Red
Fake GPS app	Mock-location rejection where exposed; travel-speed check	Send a mock flag; expect Red

Attempt	Our defence	How we test it
Double-tapping or two phones at once	LockService, active-visit check, request IDs	Fire repeated requests; expect exactly one visit (AT-04)
Calling an admin function directly	Server-side role check on every endpoint, including doPost	Call each endpoint as each role; expect denial (AT-07, AT-11)
Borrowing without being in the lab	Verified CHECKED_IN gate on the server	Request while checked out, and while pending review; expect rejection (AT-06)
Equipment never returned	Due dates, overdue flag, reminders, audit trail	Create a loan with a past due date (AT-08)
Reading phone numbers from the sheet	Encrypted at rest (enc:v1), masked in most screens, tamper detected	Inspect the Users sheet; edit a cell and expect a “damaged” warning
Private data in an export	Field minimisation (no mobile numbers) and role-restricted exports	Check export contents per role

Privacy. Raw coordinates are never stored or logged. Public inventory views never show borrower names or phone numbers. Exports are limited by role, and retention periods will be agreed with the institution.

Known limitations. GPS can be wrong indoors, browser mock-location detection is patchy, and software alone cannot stop a determined person who is physically present. LabPulse makes misuse harder, visible and auditable; it does not claim to make it impossible.

9. Non-Functional Requirements

Area	What we commit to
Speed	Ordinary actions feel quick; our test target is about 3 seconds under the load agreed with the supervisor. This is a target to measure, not a Google guarantee.
Reliability	Retries never create duplicates. Occupancy and loan states can always be rebuilt from stored records.
Ease of use	A first-time student can check in without instructions. Status colours are consistent and always paired with a word.
Compatibility	Current Chrome, Edge, Safari and Firefox on phone and laptop, with location enabled.
Scalability	Load-test against Apps Script and Sheets quotas. If limits are not enough, document it and ask for approval before changing the platform.
Maintainability	Modular code, named constants, documented sheet columns, version control, automated tests, a nightly backup copy of the spreadsheet.

10. Testing and Demo Plan

10.1 Acceptance tests

Each acceptance test is automated. The tests run the real server code against in-memory stand-ins for Sheets, CacheService, LockService, PropertiesService and Session, with a controllable clock. Results as of 18 October 2026:

ID	Test	Expected	Result
AT-01	Sign in with an outside Gmail account	Rejected	Pass

ID	Test	Expected	Result
AT-02	Valid QR, inside radius	Green check-in	Pass
AT-03	Expired, malformed and reused QR	All rejected	Pass
AT-04	Two simultaneous check-ins by one user	One active visit	Pass
AT-05	Check in and out repeatedly	Occupancy always correct	Pass
AT-06	Booking request while checked out or pending review	Blocked	Pass
AT-07	Student tries to approve a loan	Denied	Pass
AT-08	Loan past its due date	Overdue, reminded, in the digest	Pass
AT-09	Reports against known test data	Totals match	Pass
AT-10	PDF generation and daily email	PDF in Drive, email delivered	Logic passes; real Docs/Gmail to verify after deployment
AT-11	Unauthorised role change or export	Denied and logged	Pass
AT-12	Inspect all sheets after check-in	No raw coordinates anywhere	Pass
AT-13	Poor GPS accuracy	Amber, shown as pending, in review queue; counted after confirm	Pass
AT-14	Forget to check out	Visit auto-closes and is labelled	Pass
AT-15	Check in outside operating hours	Rejected; staff override is logged	Pass

In total **68 automated checks** pass: **37 server tests** (the 15 above plus token forgery, travel speed, idempotent retries, reconciliation, role overrides, unlisted sessions, Excel export, doPost and export minimisation), **18 tests for the supervisor's briefing** (roles, HOD read-only, systems and projects, the reservation queue and no-shows, lookups, rollups and archiving, encryption and tamper detection, validation, the missing list) and **13 UI tests** (every dashboard for every role, phone check-in flows, reservations, lookups, the handshake and the lab monitor, rendered against the real server code).

10.2 Still to verify on the deployed app

- Real Google Docs to PDF, Drive files, Excel export and Gmail delivery (AT-10).
- GPS and Google sign-in inside the Apps Script frame on real Android phones and iPhones.
- Identity for both @karunya.edu and @karunya.edu.in under the chosen deployment (Section 13, question 1).
- Apps Script quota use with a full class arriving at once.

10.3 Demonstration scenario

1. The monitor shows the QR. A student scans and checks in; the counter goes up on the faculty screen.
2. Re-scan the same QR on a second phone. It is rejected and logged; the monitor already shows a new code.
3. A student with poor GPS lands in Amber and appears as "pending" beside the counter; faculty confirm with one tap and the counter goes up.

4 A student whose course is not in the timetable types its code, name and hours; lab staff confirm it.

5 The checked-in student requests a Raspberry Pi; staff issue it and the Lab Map updates. A past-due loan shows as overdue.

6 Generate the daily PDF report and an Excel export.

11. Plan to the Deadline

When	Focus	Status
14 Oct	Submit SRS v1.2. Apps Script project, sheets and sign-in.	Done
15–16 Oct	Roles, settings, QR generation and validation.	Done
17–18 Oct	Geofence, check-in and check-out, occupancy, confidence scoring. Align with the client proposal (v1.3).	Done
19 Oct	Supervisor briefing: Coordinator and HOD roles, systems and projects, reservation queue, lookups, archiving, encryption, missing list (v1.4).	Done
20 Oct	Deploy to Google. Test on real phones in the lab; tune radius and accuracy thresholds.	Next
21 Oct	Verify PDF, Excel, email and triggers on the deployment; load test.	Planned
22 Oct	Demo data, bug fixes. Review-ready.	Planned
23 Oct	Freeze at 5 PM. Final fixes only. Submission.	Planned

Because the build is ahead of plan, the remaining days go to deployment and real-device testing, which is where the remaining risk lies. Rule: P0 first, then P1, then P2.

12. Risks and Limitations

Risk	What could happen	What we do
Apps Script quotas and speed	A busy class slows check-ins	Locks, batched reads and writes, cache-only monitor, a load test, documented limits
Indoor GPS drift	Honest students rejected	Amber review path and tunable thresholds
Single-use QR throughput	Queue at the door	Consume-and-Refresh, measured; fallbacks need approval
Location or sign-in behaves differently on some phones (sandboxed frame)	Check-in fails for some users	Early testing on Android and iPhone; staff-assisted check-in as fallback
Two Google Workspace domains	One domain cannot be identified	Confirm with IT before the demo (question 1)
Monitor or network outage	Nobody can check in	Visible offline state and a manual fallback; never silent attendance

13. Questions for the Supervisor

1 Are @karunya.edu and @karunya.edu.in in the same Google Workspace? The deployed app can identify users only from the deploying account’s Workspace.

2 What are the exact lab coordinates and radius, and is staff-assisted manual entry the accepted fallback when a student refuses location permission?

3 Must students always check out? Which attendance corrections need approval beyond the audit log?

4 What are the booking windows, loan duration, reminder schedule and rules for damaged items?

5 Who receives reports, how long is data kept, and who can see exports?

6 How many users and labs, and what peak arrival rate should we design for?

7 Are the added ideas (Presence Confidence, Lab Map, Lab Passport) part of the assessed scope, or extras?

8 Is a read-only HOD view the intended scope, and who should hold the Coordinator role for the Intel Unnati IoT Lab?

9 How long should archived years and backups be kept (defaults: archive after 3 months, keep 14 daily backups plus one per month)?

Status statement: all requirements in this document are implemented and pass automated tests on local stand-ins for Google services. The system has not yet been deployed to Google Apps Script or tested on real phones; Section 10.2 lists what remains to verify.